Lecture 27: Vector-Valued Functions

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define vector-valued functions and explain their use in representing curves in space.
- ☐ Evaluate limits and test for continuity of vector functions.
- ☐ Differentiate vector functions and interpret the derivative as the tangent vector to the curve.
- ☐ Find the velocity, speed, and acceleration of a particle moving along a curve in space.
- ☐ Apply differentiation rules for vector functions.
- ☐ Find the integrals of vector functions.

Vector Functions

A vector function (or vector-valued function) is a function that takes variables and returns a vector.

$$\boldsymbol{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}.$$

When a particle moves through space during a time interval I, we think of the particle's coordinates as functions defined on I

$$x = f(t), \quad y = g(t), \quad z = h(t), \quad t \in I.$$

Vector Functions

In general, when considering the domain of a vector function, we take the common domain of its components, i.e., the set of all t's where all the component functions are defined.

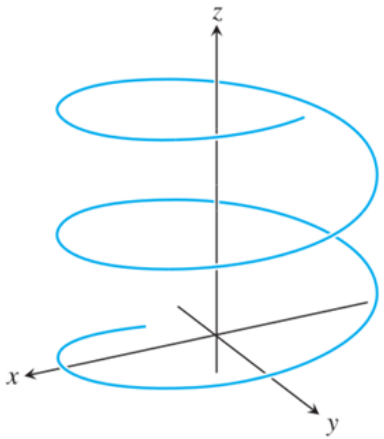
Example: Determine the domain of the vector function

$$\boldsymbol{r}(t) = \langle e^t, \ln(3-t), \sqrt{t} \rangle.$$

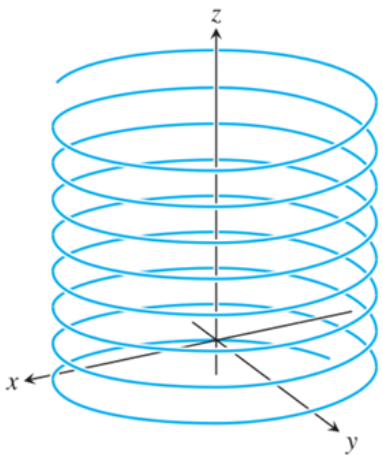
Vector Functions

Example: Graph the vector function

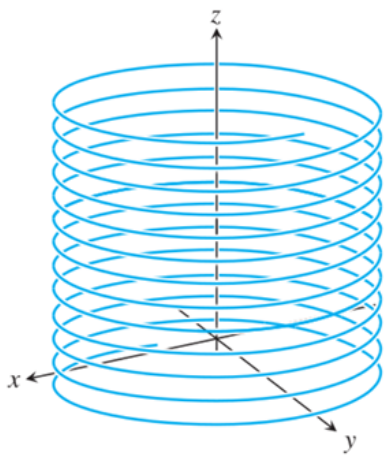
$$\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + t\boldsymbol{k}.$$



$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$



$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + 0.3t\mathbf{k}$



$\mathbf{r}(t) = (\cos 5t)\mathbf{i} + (\sin 5t)\mathbf{j} + t\mathbf{k}$

Limits

Let $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ be a vector function with domain D, and let $\boldsymbol{L}$ be a vector. We say $\boldsymbol{r}$ has limit $\boldsymbol{L}$ as t approaches t_0 and write

$$\lim_{t \to t_0} \boldsymbol{r}(t) = \boldsymbol{L}$$

if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that, for all $t \in D$,

$$|\boldsymbol{r}(t) - \boldsymbol{L}| < \epsilon \quad \text{whenever} \quad 0 < |t - t_0| < \delta.$$

If $\boldsymbol{L} = L_1\boldsymbol{i} + L_2\boldsymbol{j} + L_3\boldsymbol{k}$, then $\lim_{t \to t_0} \boldsymbol{r}(t) = \boldsymbol{L}$ if

$$\lim_{t \to t_0} f(t) = L_1, \quad \lim_{t \to t_0} g(t) = L_2, \quad \text{and} \quad \lim_{t \to t_0} h(t) = L_3.$$

Limits

Example: Find $\lim_{t \to \pi/4} \boldsymbol{r}(t)$ if $\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + t\boldsymbol{k}$.

Continuity

A vector function $\boldsymbol{r}(t)$ is continuous at a point $t = t_0$ in its domain if $\lim_{t \to t_0} \boldsymbol{r}(t) = \boldsymbol{r}(t_0)$. The function is continuous if it is continuous at every point in its domain.

$\boldsymbol{r}(t)$ is continuous at $t = t_0$ if and only if each component function is continuous at $t = t_0$.

Example:
The function $\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + t\boldsymbol{k}$ is continuous for all $t \in \mathbb{R}$.

The function $\boldsymbol{r}(t) = (\cos t)\boldsymbol{i} + (\sin t)\boldsymbol{j} + \lfloor t \rfloor \boldsymbol{k}$ is discontinuous at every integer.

Derivatives and Motion

Let $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ be the position vector of a particle moving along a curve in space.

Suppose that f, g, and h are differentiable functions of t.

The difference between the particle's positions at time t and the time $t + \Delta t$ is the vector

$$\Delta \boldsymbol{r} = \boldsymbol{r}(t + \Delta t) - \boldsymbol{r}(t)$$

Derivatives and Motion

The vector function $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ has a derivative (is differentiable) at t if f, g, and h have derivatives at t. The derivative is the vector function

$$\boldsymbol{r}'(t) = \frac{d\boldsymbol{r}}{dt} = \lim_{\Delta t \to 0} \frac{\boldsymbol{r}(t + \Delta t) - \boldsymbol{r}(t)}{\Delta t} = \frac{df}{dt}\boldsymbol{i} + \frac{dg}{dt}\boldsymbol{j} + \frac{dh}{dt}\boldsymbol{k}.$$

Example: Find the derivative of $\boldsymbol{r}(t) = 3t^2\boldsymbol{i} + e^{2t}\boldsymbol{j} + \ln(t+1)\boldsymbol{k}$.

Derivatives and Motion

A vector function is differentiable if it is differentiable at every point of its domain.

A curve is $\boldsymbol{r}$ smooth if $\frac{d\boldsymbol{r}}{dt}$ is continuous and never $\mathbf{0}$.

On a smooth curve, if $\boldsymbol{r}(t)$ is the position vector of the point P, then $\boldsymbol{r}'(t)$ is the vector tangent to the curve at P.

The tangent line to the curve at $P_0(f(t_0), g(t_0), h(t_0))$ is the line through the point P_0 parallel to the vector $\boldsymbol{r}'(t_0)$.

Derivatives and Motion

Example: Let $\boldsymbol{r}(t) = \langle \cos t, \sin t, t \rangle$ be a smooth curve. Find the equation of the tangent line at $t = \frac{\pi}{4}$.

Derivatives and Motion

Let $\boldsymbol{r}$ be the position vector of a particle moving along a smooth curve in space.

1. Velocity is the derivative of position: $\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt}$.
2. Speed is the magnitude of velocity: Speed $= |\boldsymbol{v}|$.
3. Acceleration is the derivative of velocity: $\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \frac{d^2\boldsymbol{r}}{dt^2}$.
4. The unit vector $\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$ is the direction of motion at time t.

Derivatives and Motion

Example: Find the velocity, speed, and acceleration of a particle whose motion in space is given by the position vector

$$\boldsymbol{r}(t) = 2\cos t\,\boldsymbol{i} + 2\sin t\,\boldsymbol{j} + 5\cos^2 t\,\boldsymbol{k}.$$

Derivatives and Motion

Example: Given that the position vector

$$\boldsymbol{r}(t) = 2t\cos 3t\,\boldsymbol{i} + \sin 3t\,\boldsymbol{j} + 5t\,\boldsymbol{k}.$$

Find the velocity of the particle after one full revolution.

Differentiation Rules

Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be differentiable vector functions of t, $\boldsymbol{C}$ a constant vector, c any scalar, and f any differentiable scalar function.

1. Constant Function Rule : $\frac{d}{dt}\boldsymbol{C} = \boldsymbol{0}$
2. Scalar Multiple Rules : $\frac{d}{dt}[c\boldsymbol{u}(t)] = c\boldsymbol{u}'(t)$
   $\frac{d}{dt}[f(t)\boldsymbol{u}(t)] = f'(t)\boldsymbol{u}(t) + f(t)\boldsymbol{u}'(t)$
3. Sum Rule : $\frac{d}{dt}[\boldsymbol{u}(t) + \boldsymbol{v}(t)] = \boldsymbol{u}'(t) + \boldsymbol{v}'(t)$
4. Difference Rule : $\frac{d}{dt}[\boldsymbol{u}(t) - \boldsymbol{v}(t)] = \boldsymbol{u}'(t) - \boldsymbol{v}'(t)$
5. Dot Product Rule : $\frac{d}{dt}[\boldsymbol{u}(t) \cdot \boldsymbol{v}(t)] = \boldsymbol{u}'(t) \cdot \boldsymbol{v}(t) + \boldsymbol{u}(t) \cdot \boldsymbol{v}'(t)$
6. Cross Product Rule : $\frac{d}{dt}[\boldsymbol{u}(t) \times \boldsymbol{v}(t)] = \boldsymbol{u}'(t) \times \boldsymbol{v}(t) + \boldsymbol{u}(t) \times \boldsymbol{v}'(t)$
7. Chain Rule : $\frac{d}{dt}[\boldsymbol{u}(f(t))] = f'(t)\boldsymbol{u}'(f(t))$

Vector Functions of Constant Length

When a particle is moving on a sphere centered at the origin, the position vector has a constant length equal to the radius of the sphere.

The velocity vector $\frac{d\boldsymbol{r}}{dt}$, tangent to the path of motion, is tangent to the sphere and hence perpendicular to $\boldsymbol{r}$.

For a differentiable vector function of constant length, $\boldsymbol{r}(t)$ and $\boldsymbol{r}'(t)$ are orthogonal.

Vector Functions of Constant Length

If $\boldsymbol{r}$ is a differentiable vector function of t and the length of $\boldsymbol{r}(t)$ is constant, then

$$\boldsymbol{r} \cdot \frac{d\boldsymbol{r}}{dt} = 0.$$

Example: $\boldsymbol{r}(t) = \cos t\, \boldsymbol{i} + \sin t\, \boldsymbol{j} + \boldsymbol{k}$

Integrals of Vector Functions

A differentiable vector function $\boldsymbol{R}(t)$ is an antiderivative of a vector function $\boldsymbol{r}(t)$ on an interval I if $\frac{d\boldsymbol{R}}{dt} = \boldsymbol{r}$ at each point of I.

If $\boldsymbol{R}$ is an antiderivative of $\boldsymbol{r}$ on I, then every antiderivative of $\boldsymbol{r}$ on I has the form $\boldsymbol{R} + \boldsymbol{C}$ for some constant vector $\boldsymbol{C}$.

The indefinite integral of $\boldsymbol{r}$ with respect to t is the set of all antiderivatives of $\boldsymbol{r}$, denoted by

$$\int \boldsymbol{r}(t)\, dt.$$

Example: $\int ((\cos t)\boldsymbol{i} + \boldsymbol{j} - 2t\boldsymbol{k})\, dt$

Integrals of Vector Functions

If the components of $\boldsymbol{r}(t) = f(t)\boldsymbol{i} + g(t)\boldsymbol{j} + h(t)\boldsymbol{k}$ are integrable over $[a, b]$, then so is $\boldsymbol{r}$, and the definite integral of $\boldsymbol{r}$ from a to b is

$$\int_a^b \boldsymbol{r}(t)\,dt = \left(\int_a^b f(t)\,dt\right)\boldsymbol{i} + \left(\int_a^b g(t)\,dt\right)\boldsymbol{j} + \left(\int_a^b h(t)\,dt\right)\boldsymbol{k}.$$

Example: $\displaystyle\int_0^{\pi} ((\cos t)\boldsymbol{i} + \boldsymbol{j} - 2t\boldsymbol{k})\,dt$

The Fundamental Theorem of Calculus for continuous vector functions:

$$\int_a^b \boldsymbol{r}(t)\,dt = \boldsymbol{R}(t)\Big]_a^b = \boldsymbol{R}(b) - \boldsymbol{R}(a).$$

Integrals of Vector Functions

Example: A moving particle departed from the point $(4, 0, 0)$ with initial velocity $\boldsymbol{v}(0) = 3\boldsymbol{j}$. Suppose that its acceleration is $\boldsymbol{a}(t) = -(3\cos t)\boldsymbol{i} - (3\sin t)\boldsymbol{j} + 2\boldsymbol{k}$. Find its velocity and position at time t.